

SpendSmart Plaid Production Deployment and OAuth Runbook

Internal technical documentation for rebuilding, maintaining, and troubleshooting the SpendSmart Plaid integration infrastructure.

This document explains what was configured, why each component exists, how the pieces communicate, and what is required to move from Sandbox testing to Production usage.

1. Overview - Why Plaid Exists in SpendSmart

SpendSmart uses Plaid to securely connect users' financial institutions. The app does not directly store bank credentials or communicate with banks.

Plaid acts as the secure intermediary:

- SpendSmart requests a Plaid Link session.
- The user authenticates with their financial institution.
- Plaid securely exchanges authorization information.
- SpendSmart receives approved account and transaction data through the backend.

OAuth is required by many financial institutions because the bank controls authentication. The user temporarily leaves SpendSmart, authenticates with the bank, and returns through a secure redirect process.

2. Complete Architecture

The final architecture contains several coordinated systems:

iPhone App
|
SpendSmart Backend (Supabase Edge Functions)
|
Plaid API
|
Financial Institution

OAuth return path:

Bank Authentication
|
<https://plaid.sldevapps.com/spendsmart/plaid/>
|
Cloudflare Pages
|
Apple Universal Links
|
SpendSmart iPhone App

Each component has a specific responsibility and all redirect values must match exactly.

3. Network Solutions DNS Configuration

Network Solutions manages DNS for the sldevapps.com domain.

A dedicated subdomain was created:

plaid.sldevapps.com

Purpose:

This isolates Plaid OAuth traffic from the main website and allows Apple and Plaid to use a dedicated secure endpoint.

DNS record created:

Type:

CNAME

Host:

plaid

Target:

spendsmart-plaid-oauth-pages.pages.dev

This creates:

plaid.sldevapps.com -> Cloudflare Pages deployment

Verification command:

dig plaid.sldevapps.com CNAME

A successful result confirms DNS points to Cloudflare.

4. Cloudflare Pages Configuration

Cloudflare Pages hosts the Apple App Site Association file.

Apple requires apps using Universal Links to prove ownership of a website by checking:

<https://domain.com/.well-known/apple-app-site-association>

Cloudflare provides:

- HTTPS hosting
- Valid SSL certificate
- Reliable global delivery

Hosted project:

spendsmart-plaid-oauth-pages

Custom domain:

plaid.sldevapps.com

Required folder structure:

.well-known/
apple-app-site-association

The filename has no extension. It is not .json or .txt.

The final hosted URL is:

<https://plaid.sldevapps.com/.well-known/apple-app-site-association>

5. Apple Universal Links

Universal Links allow iOS to recognize that a web address belongs to the SpendSmart app.

Without Universal Links:
The OAuth return opens Safari.

With Universal Links:
iOS recognizes ownership and returns the user directly to SpendSmart.

Xcode Associated Domain:

applinks:plaid.sldevapps.com

The Apple App Site Association file contains:

Team ID:
2UZJX3TEG4

Bundle ID:
com.scott.financetrack

Combined App Identifier:

2UZJX3TEG4.com.scott.financetrack

Supported paths:

/spendsmart/plaid
/spendsmart/plaid/*

6. Plaid Dashboard Configuration

Plaid requires an Allowed Redirect URI so it knows where OAuth authentication should return.

Configured value:

<https://plaid.sldevapps.com/spendsmart/plaid/>

The trailing slash is important.

This value must match:

- Plaid Dashboard
- SpendSmart backend
- OAuth configuration

A mismatch causes OAuth failures.

7. SpendSmart Backend Configuration

The backend contains the same redirect URI:

<https://plaid.sldevapps.com/spendsmart/plaid/>

It is used when creating Plaid Link sessions.

Both supported flows use this:

- New bank connection
- Existing connection reconnect/update mode

The backend is responsible for communicating with Plaid. Secrets and access tokens remain server-side and are never placed in the iPhone app.

8. Plaid Production Requirements

Before production usage, Plaid requires completion of their production onboarding process.

Typical requirements include:

Account and organization:

- Business information
- Contact information
- Support information

Application information:

- App description
- Website
- Privacy policy
- Terms of service

Products:

- Transactions product approval
- Other requested Plaid products if needed

Compliance:

- Data usage disclosures
- User consent flows

- Security practices
- Attestations

Security requirements:

- Plaid secret stored only on backend
- HTTPS endpoints
- No credentials exposed in the mobile application

9. Sandbox to Production Migration

Do not switch environments until Production approval and credentials are confirmed.

Current Sandbox process:

PLAID_ENV=sandbox

Future Production process:

PLAID_ENV=production

Migration steps:

1. Confirm Plaid Production approval.
2. Obtain Production client ID and secret.
3. Update Supabase secrets.
4. Deploy Edge Functions.
5. Verify backend logs.
6. Perform real institution testing.

Never delete Sandbox data before Production configuration is verified.

10. Sandbox Cleanup Strategy

When moving to Production, remove only Sandbox-related financial connection data.

Remove:

- Sandbox Plaid items
- Sandbox linked accounts
- Sandbox imported transactions
- Local Sandbox connection state

Preserve:

- Manual accounts
- Manual transactions
- Budgets
- Monthly plans
- User settings
- Spend Sense configuration

11. Production Validation Checklist

After Production activation:

Test:

- ☐ Connect a real institution
- ☐ Complete OAuth login
- ☐ Return successfully to SpendSmart
- ☐ Accounts appear correctly
- ☐ Balances synchronize
- ☐ Transactions import
- ☐ Pending transactions behave correctly
- ☐ Refresh works
- ☐ Disconnect works
- ☐ Reconnect works

Validate:

- No duplicate transactions
- Correct account ownership
- Correct user isolation

12. Troubleshooting Guide

AASA not working:

Check:

<https://plaid.sldevapps.com/.well-known/apple-app-site-association>

Verify:

- HTTP 200 response
- application/json content type
- Correct App ID

OAuth opens Safari instead of app:

Check:

- Associated Domains entitlement
- AASA file
- Domain spelling

Plaid invalid redirect URI:

Check exact match including trailing slash.

Production API errors:

Check:

- Production credentials
- PLAID_ENV
- Edge Function deployment
- Plaid approval status

13. Maintenance Rules

If the Plaid OAuth domain changes, update every location:

1. Network Solutions DNS
2. Cloudflare Pages custom domain
3. Apple App Site Association file hosting
4. Plaid Allowed Redirect URI
5. Xcode Associated Domains
6. SpendSmart backend redirect URI

All six values must remain synchronized.

Final Status

The SpendSmart Plaid OAuth infrastructure foundation is complete.

Completed:

- DNS
- Cloudflare hosting
- Apple Universal Links
- Plaid redirect registration
- Xcode configuration
- Backend configuration

Remaining:

- Plaid Production onboarding completion
- Production credentials
- Backend production deployment
- Real institution testing